Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of $1 - \cot 23^\circ$ equals

(A) $\frac{2}{1+\cot 22^\circ}$

(B) $\frac{1}{1+\cot 22^\circ}$

(C) $\frac{2}{1-\cot 22^\circ}$

(D) $\frac{1}{1-\cot 22^\circ}$

Q.2

If $\sin\theta = \frac{x^2+y^2}{x^2-y^2}$, where $x, y \in R$ then

(A) $x \neq y$ where x and $y \in R$

(B) $x \neq y$ and $x, y \in R+$

(C) $x = 0$ or $y = 0$

(D) $x \neq y$ and $x \neq 0, y \neq 0$

Q.3

If $\sin A + \sin B = a$ and $\cos A + \cos B = b$, then $\cos(A + B)$ is

(A) $\frac{a^2+b^2}{b^2-a^2}$

(B) $\frac{2ab}{a^2+b^2}$

(C) $\frac{b^2-a^2}{a^2+b^2}$

(D) $\frac{a^2-b^2}{a^2+b^2}$

Q.4

The value of $\cos(\alpha+\beta+\gamma) + \cos(\gamma+\alpha-\beta) + \cos(\alpha+\beta-\gamma) + \cos(\beta+\gamma-\alpha)$ is

(A) $4\cos\alpha \cdot \cos\beta \cdot \cos\gamma$

(B) $2\cos\alpha \cdot \cos\beta \cdot \cos\gamma$

(C) $2\sin\alpha \cdot \cos\beta \cdot \sin\gamma$

(D) $4\sin\alpha \cdot \sin\beta \cdot \sin\gamma$

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Q.5

Given that $\cos\left(\frac{\alpha-\beta}{2}\right)=2\cos\left(\frac{\alpha+\beta}{2}\right)$, then $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{1}{8}$

Q.6

Let $f(\theta)=\tan\left(\frac{\pi}{3}+\theta\right)\tan\left(\frac{\pi}{3}-\theta\right), -\frac{\pi}{3}<\theta<\frac{\pi}{3}$, then $f\left(\frac{\pi}{12}\right)$ is equal to:

(A) $2+\sqrt{3}$

(B) $2-\sqrt{3}$

(C) $\sqrt{3}+1$

(D) $\sqrt{3}-1$

Q.7

If $\operatorname{cosec}\theta=\frac{p+q}{p-q}$ where $p>q>0$, then $\left|\cot\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right|$ is equal to:

(A) $\sqrt{\frac{p}{q}}$

(B) $\sqrt{\frac{q}{p}}$

(C) $\sqrt{pq}$

(D) pq

Q.8

$\sin^2\alpha+\cos^2(\alpha+\beta)+2\sin\alpha\sin\beta\cos(\alpha+\beta)$ is independent of

(A) α

(B) β

(C) both α and β

(D) none

Q.9

The value of $\cos 12^\circ+\cos 84^\circ+\cos 156^\circ+\cos 132^\circ$ is

(A) $1/8$

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(B) $-1/2$

(C) 1

(D) $1/2$

Q.10

If $\sin \mathrm{A} = 3 \sin(\mathrm{A} + 2\ \mathrm{B})$, angle B is acute and A is obtuse; then

(A) $\tan \mathrm{B} = 1/\sqrt{2}$

(B) $\tan \mathrm{B} > 1/\sqrt{2}$

(C) $\tan \mathrm{B} < 1/\sqrt{2}$

(D) $0 < \tan \mathrm{B} < 1/\sqrt{2}$

Q.11

If $\cos \alpha + \cos \beta = a, \sin \alpha + \sin \beta = b$ and $\alpha - \beta = 2\theta$, then $\frac{\cos 3\theta}{\cos \theta} =$

(A) $a^2 + b^2 - 2$

(B) $a^2 + b^2 - 3$

(C) $3 - a^2 - b^2$

(D) $\left(a^2 + b^2\right)/4$

Q.12

If $\cos(\alpha + \beta) = 4/5$ and $\sin(\alpha - \beta) = 5/13$ where $0 \leq \alpha, \beta \leq \pi/4$, then $\tan 2\alpha =$

(A) $19/12$

(B) $20/7$

(C) $25/16$

(D) $56/33$

Q.13

The value of $\log_3 \tan 1^\circ + \log_3 \tan 2^\circ + \ldots + \log_3 \tan 89^\circ$ is

(A) 3

(B) 1

(C) 2

(D) 0

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Q.14

If $u = \sqrt{a^2 \cos^2\theta + b^2 \sin^2\theta} + \sqrt{a^2 \sin^2\theta + b^2 \cos^2\theta}$, then the difference between the maximum and minimum values of u^2 is given by

(A) $(a+b)^2$

(B) $2\sqrt{a^2+b^2}$

(C) $2\left(a^2+b^2\right)$

(D) $(a-b)^2$

Q.15

$\sin x, \sin 2x, \sin 3x$ are in A.P. if

(A) $x = 165^\circ$

(B) $x = 195^\circ$

(C) $x = 15^\circ$

(D) $x = 2n\pi \quad n \in I$

Q.16

Let $f(\theta) = \sin\theta \sin 3\theta \sin 5\theta$, then $f(\pi/14)$ is equal to:

(A) $1/8$

(B) $1/4$

(C) $1/7$

(D) $1/14$

Q.17

Which of the following is correct?

(A) $\sin 1^\circ > \sin 1$

(B) $\sin 1^\circ < \sin 1$

(C) $\sin 1^\circ = \sin 1$

(D) $\sin 1^\circ = \frac{\pi}{180} \sin 1$

Q.18

If $x_{n+1} = \sqrt{\frac{1}{2}(1+x_n)}$, then $\cos\left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 x_3 \ldots \text{ to infinite}}\right] (-1 < x_0 < 1)$ is equal to

(A) -1

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(B) 1

(C) x_0

(D) $\frac{1}{x_0}$

Q.19

If A, B, C are the angles of a triangle such that angle A is obtuse then

(A) $\tan A \tan B < 1$

(B) $\tan B \tan C < 1$

(C) $\tan C \tan A < 1$

(D) $\tan A \tan B \tan C < 1$

Q.20

If $n \in \mathbf{N}$ and $\sin\left(\frac{\pi}{2n}\right) + \cos\left(\frac{\pi}{2n}\right) = \frac{\sqrt{n}}{2}$, then a possible value of n is

(A) 4

(B) 6

(C) 8

(D) 12

Q.21

If $\sin x + \sin^2 x + \sin^3 x = 1$, then $\cos^6 x - 4\cos^4 x + 8\cos^2 x =$

Q.22

If $\sin x + \operatorname{cosec} x = 2$, then $\sin^n x + \operatorname{cosec}^n x$ is equal to

Q.23

The least integer less than or equal to the minimum value of $f(x) = |\sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x|$

Q.24

If $x = \cot\theta + \tan\theta; y = \sec\theta - \cos\theta$ then $\left(x^{4/3}y^{2/3} - x^{2/3}y^{4/3}\right)$

Q.25

If $\sin^2 A = x$, then $\sin A \sin 2A \sin 3A \sin 4A$ is a polynomial in x, the sum of whose coefficients is

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Q.26

The number of pairs (x, y) satisfying the equation $\sin x + \sin y = \sin(x + y)$ and $|x| + |y| = 1$ is

Q.27

If $\sin A, \cos A$ and $\tan A$ are in G.P., then $\cot^6 A - \cot^2 A =$

Q.28

The sides of a triangle are $\sin \beta, \cos \beta$ and $\sqrt{1 + a \sin 2\beta}$ for some $a > 0, 0 < \beta < \frac{\pi}{2}$. If the greatest angle of this triangle is 120°, then a equals (determine the value of 16a)

Q.29

$\sin\theta \cdot \sec 3\theta + \sin 3\theta \cdot \sec 3^2\theta + \sin 3^2\theta \cdot \sec 3^3\theta + \ldots$ upto n terms $=$
$\frac{1}{2}\left[\tan K^n\theta - \tan\theta\right]$, find K

Q.30

The sum of the roots of the equation $4\cos^3 x - 4\cos^2 x - \cos(\pi + x) - 1 = 0$ in the interval $[0, 315]$ is $p\pi$, where p is equal to

Answers & Solutions

Q.1 Answer:

$\frac{2}{1-\cot 22^\circ}$

Solution:

$45^\circ = 22^\circ + 23^\circ$. **Use**

$\cot(A + B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$

Q.2 Answer:

$x = 0$ or $y = 0$

Solution:

$-1 \le \frac{x^2 + y^2}{x^2 - y^2} \le 1$

(since $|\sin\theta| \le 1$**)**

$\therefore \quad x = 0$ **or** $y = 0$

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Q.3 Answer:

$\frac{b^2-a^2}{a^2+b^2}$

Solution:

$a^2 = \sin^2 A + \sin^2 B + 2\sin A \sin B$

$b^2 = \cos^2 A + \cos^2 B + 2\cos A \cos B$

$\left(a^2 + b^2\right) = 2 + 2(\cos A \cos B + \sin A \sin B)$

$= 2 + 2\cos(A - B)$

$\left(b^2 - a^2\right) = \cos 2A + \cos 2B + 2(\cos A \cos B) - 2\sin A \sin B$

$= \cos 2A + \cos 2B + 2\cos(A + B) = 2\cos(A + B)\cos(A - B) + 2\cos(A + B)$

$\left(b^2 - a^2\right) = 2\cos(A + B)[\cos(A - B) + 1]$

$\left[b^2 - a^2\right) = 2\cos(A + B)\left[\frac{\left(a^2 + b^2 - 2\right)}{2} + 1\right] = 2\cos(A + B)$

$\cos(A + B) = \frac{\left(b^2-a^2\right)}{(b^2+a^2)}$

Q.4 Answer:

$4\cos\alpha \cdot \cos\beta \cdot \cos\gamma$

Solution:

Given expression can be rearranged as

$$\cos(\alpha + \beta + \gamma) + \cos(\alpha + \beta - \gamma) + \cos(\gamma + \alpha - \beta) + \cos(\beta + \gamma - \alpha)$$
$$= 2\cos\gamma \cdot \cos(\alpha + \beta) + 2\cos\gamma\cos(\alpha - \beta)$$
$$= 2\cos\gamma(2\cos\alpha \cdot \cos\beta)$$
$$= 4\cos\alpha \cdot \cos\beta \cdot \cos\gamma$$

Q.5 Answer:

$\frac{1}{3}$

Solution:

$\cos\left(\frac{\alpha-\beta}{2}\right) = 2\cos\left(\frac{\alpha+\beta}{2}\right) \Rightarrow$

$\cos\frac{\alpha}{2}\cos\frac{\beta}{2} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2} = 2\cos\frac{\alpha}{2}\cos\frac{\beta}{2} - 2\sin\frac{\alpha}{2}\sin\frac{\beta}{2}$

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$$\Rightarrow \quad 3\sin\frac{\alpha}{2}\sin\frac{\beta}{2}=\cos\frac{\alpha}{2}\cos\frac{\beta}{2}\Rightarrow\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{1}{3}$$

Q.6 Answer:

$2+\sqrt{3}$

Solution:

$$f(\theta)=\frac{\sin\left(\frac{\pi}{3}+\theta\right)\sin\left(\frac{\pi}{3}-\theta\right)}{\cos\left(\frac{\pi}{3}+\theta\right)\cos\left(\frac{\pi}{3}-\theta\right)}$$

$$=\frac{\sin^2\left(\frac{\pi}{3}\right)-\sin^2\theta}{\cos^2\left(\frac{\pi}{3}\right)-\sin^2\theta}$$

$$=\frac{3-4\sin^2\theta}{1-4\sin^2\theta}=\frac{2\cos(2\theta)+1}{2\cos(2\theta)-1}$$

$$\therefore \quad f\left(\frac{\pi}{12}\right)=\frac{2\cos\left(\frac{\pi}{6}\right)+1}{2\cos\left(\frac{\pi}{6}\right)-1}=\frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$=2+\sqrt{3}$$

Q.7 Answer:

$\sqrt{\frac{q}{p}}$

Solution:

$$\operatorname{cosec}\theta=\frac{p+q}{p-q}$$

$$\Rightarrow \frac{1}{\sin\theta}=\frac{p+q}{p-q}$$

Applying componendo and dividendo

$$\Rightarrow \frac{1+\sin\theta}{1-\sin\theta}=\frac{p+q+p-q}{p+q-p+q}$$

$$\Rightarrow \left(\frac{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}\right)^2=\frac{p}{q}$$

$$\Rightarrow \left(\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}\right)^2=\frac{p}{q}$$

$$\Rightarrow \tan^2\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=\frac{\mathrm{p}}{\mathrm{q}}$$

$$\Rightarrow \cot^2\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=\frac{\mathrm{q}}{\mathrm{p}}$$

$$\Rightarrow \cot\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=\sqrt{\frac{\mathrm{q}}{\mathrm{p}}}$$

Q.8 Answer:

α

Solution:

The given expression is equal to

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$$\sin^2\alpha+\cos(\alpha+\beta)[\cos(\alpha+\beta)+2\sin\alpha\sin\beta]$$
$$=\sin^2\alpha+\cos(\alpha+\beta)[\cos\alpha\cos\beta+\sin\alpha\sin\beta]$$
$$=\sin^2\alpha+\cos(\alpha+\beta)\cos(\alpha-\beta)$$
$$=\sin^2\alpha+\cos^2\alpha-\sin^2\beta=1-\sin^2\beta=\cos^2\beta$$

which is independent of α only.

Q.9 Answer:

$-1/2$

Solution:

$(\cos 12^\circ+\cos 132^\circ)+(\cos 84^\circ+\cos 156^\circ)=2\cos\frac{12^\circ+132^\circ}{2}\cos\frac{12^\circ-132^\circ}{2}+$
$2\cos\frac{84^\circ+156^\circ}{2}\cos\frac{84^\circ-156^\circ}{2}=2\cos 72^\circ\cos 60^\circ+2\cos 120^\circ\cos 36^\circ=2\times\frac{\sqrt{5}-1}{4}\times\frac{1}{2}+$
$2\times\left(-\frac{1}{2}\right)\times\frac{\sqrt{5}+1}{4}=-\frac{1}{2}$

Q.10 Answer:

$0<\tan B<1/\sqrt{2}$

Solution:

$\frac{\sin(A+2B)}{\sin A}=\frac{1}{3}$

By componendo and dividendo, $\tan(A+B)=-2\tan B$

$\Rightarrow \quad \frac{\tan A+\tan B}{1-\tan A\tan B}=-2\tan B$

$\Rightarrow \quad \tan A=\frac{3\tan B}{2\tan^2 B-1}<0 \quad (\text{as A is obtuse})$
$\Rightarrow \quad 2\tan^2 B-1<0(\text{ as } \tan B>0)$
$\Rightarrow \quad 0<\tan B<1/\sqrt{2}$

Q.11 Answer:

a^2+b^2-3

Solution:

From the given relations we have

$$a^2+b^2=\cos^2\alpha+\cos^2\beta+2\cos\alpha\cos\beta+\sin^2\alpha+$$
$$\sin^2\beta+2\sin\alpha\sin\beta$$
$$=2+2\cos(\alpha-\beta)$$
$$=2+2\cos 2\theta=4\cos^2\theta$$

Now $\frac{\cos 3\theta}{\cos\theta}=\frac{4\cos^3\theta-3\cos\theta}{\cos\theta}$

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$= 4\cos^2\theta - 3 = a^2 + b^2 - 3$

Q.12 Answer:

$56/33$

Solution:

$0 \le \alpha, \beta \le \pi/4$

$\Rightarrow \quad 0 \le \alpha + \beta \le \pi/2 \quad \Rightarrow \quad -\pi/4 \le \alpha - \beta \le \pi/4$

Now $\cos(\alpha+\beta) = 4/5 \Rightarrow \tan(\alpha+\beta) = 3/4$

and $\sin(\alpha-\beta) = 5/13 \Rightarrow \tan(\alpha-\beta) = 5/12$

we have $\tan 2\alpha = \tan[(\alpha+\beta)+(\alpha-\beta)]$

$= \frac{\tan(\alpha+\beta)+\tan(\alpha-\beta)}{1-\tan(\alpha+\beta)\tan(\alpha-\beta)}$
$= \frac{(3/4)+(5/12)}{1-(3/4)(5/12)} = \frac{14/12}{33/48} = \frac{56}{33}$

Q.13 Answer:

0

Solution:

$\log_3 \tan 1 + \log_3 \tan 2 + \cdots \log_3 \tan 89 \log_3(\tan 1 \cdot \tan 2 \cdots \tan 89)$
$(\log_a b + \log_a c = \log_a bc)$

$\log_3(\tan 1 \cdot \tan 2 \cdots \tan 45 \ldots \cot 1)$

$(\tan(90-\theta) = \cot\theta)$

$\log_3 1 = 0.$

Q.14 Answer:

$(a-b)^2$

Solution:

We have

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$$u=\sqrt{\frac{a^2}{2}(1+\cos 2\theta)+\frac{b^2}{2}(1-\cos 2\theta)}$$
$$+\sqrt{\frac{a^2}{2}(1-\cos 2\theta)+\frac{b^2}{2}(1+\cos 2\theta)}$$
$$=\sqrt{\frac{a^2+b^2}{2}+\frac{a^2-b^2}{2}\cos 2\theta}$$
$$+\sqrt{\frac{a^2+b^2}{2}-\frac{a^2-b^2}{2}\cos 2\theta}$$

Squaring we get

$$u^2=a^2+b^2+2\sqrt{\left(\tfrac{a^2+b^2}{2}\right)^2-\left(\tfrac{a^2-b^2}{2}\right)^2\cos^2 2\theta}$$

Thus $\max\left(u^2\right)=a^2+b^2+a^2+b^2=2\left(a^2+b^2\right)$

So the required difference $=2\left(a^2+b^2\right)-(a+b)^2=(a-b)^2$

Q.15 Answer:

$x=2n\pi \quad n\in I$

Solution:

$\sin x, \sin 2x, \sin 3x$ **are in A.P.**

If $2\sin 2x=\sin x+\sin 3x=2\sin 2x\cos x$

or if $2\sin 2x(1-\cos x)=0$

or if either $\sin 2x=0$ **or** $\cos x=1$

which is satisfied by $x=2n\pi$ **for which** $\cos x=1$

Q.16 Answer:

$1/8$

Solution:

As $\frac{\pi}{14}=\frac{\pi}{2}-\frac{3\pi}{7}, \frac{3\pi}{14}=\frac{\pi}{2}-\frac{2\pi}{7}$ **and** $\frac{5\pi}{14}=\frac{\pi}{2}-\frac{\pi}{7}$**, we can write**

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$$f\left(\frac{\pi}{14}\right)=\cos\frac{3\pi}{7}\cos\frac{2\pi}{7}\cos\frac{\pi}{7}$$

$$\Rightarrow 2\sin\frac{\pi}{7}f\left(\frac{\pi}{14}\right)=\sin\frac{2\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$$

$$\Rightarrow \quad 4\sin\frac{\pi}{7}f\left(\frac{\pi}{14}\right)=\sin\frac{4\pi}{7}\cos\frac{3\pi}{7}$$

$$\Rightarrow \quad 8\sin\left(\frac{\pi}{7}\right)f\left(\frac{\pi}{14}\right)=\sin\left(\frac{7\pi}{7}\right)+\sin\left(\frac{\pi}{7}\right)=\sin\frac{\pi}{7}$$

$$\Rightarrow f\left(\frac{\pi}{14}\right)=\frac{1}{8}$$

Q.17 Answer:

$\sin 1^\circ < \sin 1$

Solution:

$1^c=\frac{180^\circ}{\pi}=57.3^\circ$

$1^\circ<57.3^\circ$

$\sin 1^\circ<\sin 57.3^\circ$

$\therefore \sin 1^\circ<\sin 1^c$

Q.18 Answer:

x_0

Solution:

Let $x_0=\cos\theta$, **then** $x_1=\sqrt{\frac{1}{2}(1+\cos\theta)}=\cos\frac{\theta}{2}, x_2=\cos\left(\frac{\theta}{2^2}\right), x_3=\cos\left(\frac{\theta}{2^3}\right), \ldots..$

and so on.

so that $\left|\frac{\sqrt{1-x_0^2}}{x_1x_2x_3\ldots\ldots \text{ to infinite}}\right|$

$$=\frac{\sin\theta}{\cos\frac{\theta}{2}\cos\frac{\theta}{2^2}\ldots\ldots\cos\frac{\theta}{2^n}\ldots \text{infinite}}=\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\cos\frac{\theta}{2}\cos\frac{\theta}{2^2}\ldots\ldots\cos\frac{\theta}{2^n}\ldots\ldots \text{infinite}}$$

$$=\frac{2^2\sin\frac{\theta}{2^2}\cos\frac{\theta}{2^2}}{\cos\frac{\theta}{2^2}\cos\frac{\theta}{2^3}\ldots\ldots\cdot\cos\frac{\theta}{2^n}\ldots \text{infinite}}=\lim_{n\to\infty}\frac{2^n\sin\frac{\theta}{2^n}}{\cos\frac{\theta}{2^{n+1}}}$$

$$=\lim_{n\to\infty}\theta\left(\frac{\sin\frac{\theta}{2^n}}{\frac{\theta}{2^n}}\right)\frac{1}{\cos\frac{\theta}{2^{n+1}}}=\theta$$

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so that $\cos\left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 \ldots\ldots \text{infinite}}\right] = \cos\theta = x_0$

Q.19 Answer:

$\tan B \tan C < 1$

Solution:

$\because$ **A is obtuse angle.**

Then $\frac{\pi}{2} < \mathrm{A} < \pi$

$$\Rightarrow \frac{\pi}{2} < \pi - (B+C) < \pi$$
$$\Rightarrow -\frac{\pi}{2} > -\pi + (B+C) > -\pi$$
$$\Rightarrow \frac{\pi}{2} > (B+C) > 0$$
$$\because B + C < \frac{\pi}{2}$$
$$\Rightarrow B < \frac{\pi}{2} - C$$
$$\Rightarrow \tan B < \tan\left(\frac{\pi}{2} - C\right)$$
$$\Rightarrow \tan B < \cot C$$
$$\therefore \tan B \quad \frac{1}{\tan C}$$

Hence $\tan \mathrm{B} \tan \mathrm{C} < 1$

Q.20 Answer:

6

Solution:

$$\frac{n}{4} = \sin^2\left(\frac{\pi}{2n}\right) + \cos^2\left(\frac{\pi}{2n}\right) + \sin\left(\frac{2\pi}{2n}\right)$$
$$\Rightarrow \quad \sin\left(\frac{\pi}{n}\right) = \frac{n-4}{4}$$
$$(\pi)$$

For $n \in \mathbf{N}, \sin\left(\frac{\pi}{n}\right) > 0$

$\therefore \quad \frac{n-4}{4} > 0$ **or** $n > 4$

For $n > 4, \frac{\pi}{n} < \frac{\pi}{2} \Rightarrow \sin\left(\frac{\pi}{n}\right) < 1$

$\Rightarrow \quad 0 < \frac{1}{4}(n-4) < 1 \Rightarrow 4 < n < 8$

Thus, a possible value of $n = 6$

In fact, for $n = 6$

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$$\sin\left(\frac{\pi}{12}\right)+\cos\left(\frac{\pi}{12}\right)$$
$$=\sqrt{2}\sin\left(\frac{\pi}{4}+\frac{\pi}{12}\right)=\sqrt{2}\sin\left(\frac{\pi}{3}\right)$$
$$=\sqrt{2}\left(\frac{\sqrt{3}}{2}\right)=\frac{\sqrt{6}}{4}=\frac{\sqrt{n}}{4}$$

Q.21 Answer:

—

Solution:

Given, $1=\sin x+\sin^2 x+\sin^3 x\Rightarrow\cos^2 x=\sin x\left(1+\sin^2 x\right)=\sin x\left(2-\cos^2 x\right)$

$$\Rightarrow \quad \cos^4 x=\left(1-\cos^2 x\right)\left(4+\cos^4 x-4\cos^2 x\right)$$
$$=4-4\cos^2 x+\cos^4 x-\cos^6 x-4\cos^2 x+4\cos^4 x$$
$$\Rightarrow \quad \cos^6 x-4\cos^4 x+8\cos^2 x=4$$

Q.22 Answer:

—

Solution:

$\sin x+\operatorname{cosec} x=2\Rightarrow(\sin x-1)^2=0$

$\Rightarrow\sin x=1\Rightarrow\sin^n x+\operatorname{cosec}^n x=1+1=2$

Q.23 Answer:

—

Solution:

Let $\sin\mathrm{x}+\cos\mathrm{x}=\mathrm{t}$**, we get** $\mathrm{t}^2-1=\sin 2\mathrm{x}$

Let $\mathrm{P}=|\sin\mathrm{x}+\cos\mathrm{x}+\tan\mathrm{x}+\cot\mathrm{x}+\operatorname{cosec}\mathrm{x}+\sec\mathrm{x}|=\left|\mathrm{t}+\frac{1+\mathrm{t}}{\sin\mathrm{x}\cos\mathrm{x}}\right|=|\ \mathrm{t}+\frac{2}{t-1}\ |$

$\mathrm{t}=\sqrt{2}\sin(\mathrm{x}+45)$

If $\mathrm{p}=1-\sqrt{2}$**, then** $\mathrm{P}=2\sqrt{2}-1$

Therefore the minimum value is $2\sqrt{2}-1$

Q.24 Answer:

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Solution:

As $x\tan\theta = \sec^2\theta \quad y\sec\theta = \tan^2\theta$

$\therefore \quad x^2y = \sec^3\theta \quad$ and $\quad xy^2 = \tan^3\theta$

Now $\left(x^{4/3}y^{2/3} - x^{2/3}y^{4/3}\right) = \left(x^2y\right)^{2/3} - \left(xy^2\right)^{2/3} = \left(\sec^3\theta\right)^{2/3} - \left(\tan^3\theta\right)^{2/3} = 1$

Q.25 Answer:

—

Solution:

We have $\sin A\sin 2A\sin 3A\sin 4A$

$$
\begin{aligned}
&= \sin A(2\sin A\cos A)\left(3\sin A - 4\sin^3 A\right)\times \\
&\quad 2\sin 2A\cos 2A \\
&= 2\sin^2 A\cos A\times \sin A\left(3 - 4\sin^2 A\right)\times 2\times \\
&\quad 2\sin A\cos A\left(1 - 2\sin^2 A\right) \\
&= 8\sin^4 A\cos^2 A\left(3 - 4\sin^2 A\right)\left(1 - 2\sin^2 A\right) \\
&\quad = 8x^2(1-x)(3-4x)(1-2x) \\
&= 24x^2 - 104x^3 + 144x^4 - 64x^5
\end{aligned}
$$

The required sum $= 24 - 104 + 144 - 64 = 0.$

Q.26 Answer:

—

Solution:

$$
\begin{aligned}
&\sin x + \sin y = \sin(x+y) \\
&\Rightarrow 2\sin\tfrac{x+y}{2}\cos\tfrac{x-y}{2} = 2\sin\tfrac{x+y}{2}\cos\tfrac{x+y}{2} \\
&\Rightarrow \sin\tfrac{x+y}{2}\left(\cos\tfrac{x+y}{2} - \cos\tfrac{x-y}{2}\right) = 0 \\
&\Rightarrow \text{ either } \sin\tfrac{x+y}{2} = 0 \text{ or } \cos\tfrac{x+y}{2} = \cos\tfrac{x-y}{2}
\end{aligned}
$$

Now $\sin\frac{x+y}{2} = 0$ and $|x| + |y| = 1$

$(x, y) = \left(\frac{1}{2}, -\frac{1}{2}\right)$ or $\left(\frac{1}{2}, -\frac{1}{2}\right)$

and $\cos\frac{x+y}{2} = \cos\frac{x-y}{2}$

$\Rightarrow \sin\frac{x}{2}\sin\frac{y}{2} = 0$

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$\Rightarrow \sin\frac{x}{2} = 0$ **or** $\sin\frac{y}{2} = 0$

$\Rightarrow (x, y) = (1, 0), (-1, 0), (0, 1), (0, -1)$

So the required number of pairs is 6.

Q.27 Answer:

—

Solution:

$\cos^2 A = \sin A \tan A = \sin^2 A \cdot \sec A$

$\Rightarrow \cot^2 A = \sec A.$

$$\therefore \quad \cot^6 A - \cot^2 A = \cot^2 A\left(\cot^4 A - 1\right)$$
$$= \cot^2 A\left(\sec^2 A - 1\right)$$
$$= \cot^2 A \times \tan^2 A = 1$$

Q.28 Answer:

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Solution:

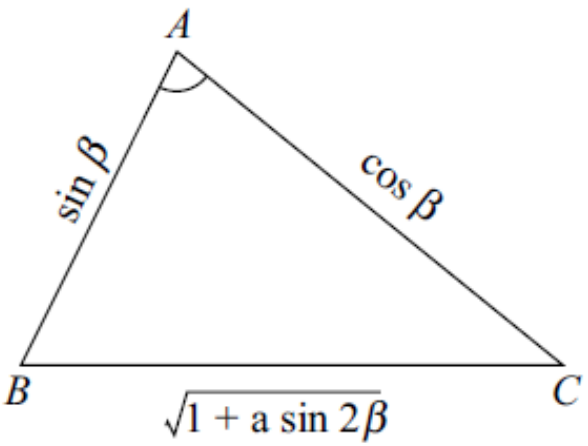
$1 + a\sin 2\beta = \sin^2\beta + \cos^2\beta - 2\sin\beta\cos\beta\cos A$

$\Rightarrow \cos A = -a$

$\Rightarrow A$ **is obtuse.**

$\Rightarrow A = 120^\circ$

$\Rightarrow a = -\cos 120^\circ = \frac{1}{2}$



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Q.29 Answer:

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Solution:

$\sin\theta \cdot \sec 3\theta + \sin 3\theta \cdot \sec 3^2\theta + \sin 3^2\theta \cdot \sec 3^3\theta + \ldots$ **upto** n **terms**

$$= \sum_{r=1}^{n} \sin 3^{r-1}\theta \cdot \sec 3^r\theta$$
$$= \sum_{r=1}^{n} \frac{2\cos 3^{r-1}\theta \sin 3^{r-1}\theta}{2\cos 3^{r-1}\theta \cdot \cos 3^r\theta}$$
$$= \frac{1}{2}\sum_{r=1}^{n} \frac{\sin\left(2.3^{r-1}\theta\right)}{\cos 3^{r-1}\theta \cdot \cos 3^r\theta}$$
$$= \frac{1}{2}\sum_{r=1}^{n} \frac{\sin\left(3^r\theta - 3^{r-1}\theta\right)}{\cos 3^{r-1}\theta \cdot \cos 3^r\theta}$$
$$= \frac{1}{2}\sum_{r=1}^{n} \frac{\sin 3^r\theta \cdot \cos 3^{r-1}\theta - \cos 3^r\theta \cdot \sin 3^{r-1}\theta}{\cos 3^{r-1}\theta \cdot \cos 3^r\theta}$$
$$= \frac{1}{2}\sum_{r=1}^{\pi} \left(\tan 3^r\theta - \tan 3^{r-1}\theta\right)$$
$$= \frac{1}{2}\left[\tan 3^n\theta - \tan\theta\right]$$

Q.30 Answer:

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Solution:

$$4\cos^3 x - 4\cos^2 x + \cos x - 1 = 0$$

$$\Rightarrow \quad \left(4\cos^2 x + 1\right)(\cos x - 1) = 0$$

$$\Rightarrow \cos x = 1 \Rightarrow x = 2n\pi$$

$$100\pi < 315 < 101\pi$$

Required sum $= 2\pi + 4\pi + \ldots 100\pi$

$$= 2(1 + 2 + \ldots 50)\pi$$

$$= \frac{2\times 50\times 51}{2}\pi = 2550\pi$$

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